Bentley, WA 6102 | 0470410719 | aalokm875@gmail.com

AALOK MANDAL

LINKEDIN PROFILE www.linkedin.com/in/aalok-mandal-a80a8536b

PROFESSIONAL PORTFOLIO akm164.github.io/My_Website

PROFESSIONAL SUMMARY

AI-focused student who has just completed a Bachelor of Computer Science (double major: Software Engineering and Artificial Intelligence & Big Data) at the University of Wollongong and is now studying a Graduate Diploma of Professional Engineering (Mining Engineering) at Curtin University, progressing to a Master of Professional Engineering (Mining Engineering) at Curtin's Kalgoorlie campus. Combines hands-on Python, machine learning and software skills with a mining engineering foundation and is currently working with Professor Erkan Topal on Smart Mine AI, a concept for AI-driven ore sorting and grinding optimisation.

Still a student, and keen to learn from experienced mining engineers while contributing practical AI skills from day one. Fully available throughout the summer break. This is the exact role I am studying and building towards: using AI to reduce manual work, improve productivity and make mining more efficient.

EDUCATION

Master of Professional Engineering **[Commencing 07/2027 – Expected Grad: 2029]**
Curtin University, Kalgoorlie, WA
Mining Engineering

Graduate Diploma of Professional Engineering **[07/2026 – Present]**
Curtin University, Bentley, WA
Mining Engineering

Bachelor of Computer Science **[07/2023 – 07/2026]**
University of Wollongong, Wollongong, NSW
Double Major: Software Engineering & Artificial Intelligence & Big Data

TECHNICAL SKILLS

- **Artificial Intelligence**
- **Python**
- **Computer Vision**
- **Automation & Prototyping**
- **Machine Learning**
- **SQL & MySQL**
- **Data Analysis & Modelling**
- **Microsoft Applications**
- **Deep Learning (CNN / ANN)**
- **MongoDB**
- **Generative AI & LLM Tools**
- **Mining Fundamentals**

SOFT SKILLS

- **Teamwork**
- **Analytical Thinking**
- **Stakeholder Collaboration**
- **Reliability**
- **Communication**
- **Adaptability**
- **Problem-Solving**
- **Leadership**
- **Attention to Detail**
- **Curiosity & Fast Learning**
- **Time Management**
- **Organisation**

WORK HISTORY

Software Engineering Assistant **[12/2025 – 07/2026]**
Realta Logic, Sydney, NSW

- Followed structured processes and checklists, tracking sprint milestones and managing backlogs in Jira to keep work on schedule and reduce delays.
- Maintained accurate technical documentation, setup guides and deployment checklists so knowledge could be shared and reused across the team.
- Executed manual test suites, logged bugs and verified fixes, applying strong attention to detail and quality assurance.
- Worked in an Agile/Scrum team, prioritising tasks and planning sprints with technical and non-technical colleagues.

- Removed blockers and improved communication across teams to keep projects moving.

Peer Tutor **[07/2024 – 11/2025]**
UOW College, Wollongong, NSW

- Provided peer technical support to students learning Python, MongoDB and MySQL.
- Diagnosed and debugged problems in student work by guiding peers step by step to the solution.
- Explained complex technical concepts in simple language so non-technical learners could understand and apply them quickly.

Chef **[04/2022 – 07/2024]**
Ariel's Café, Wollongong, NSW

- Managed stock, resources and supplier relationships, reducing waste and controlling costs.
- Led and mentored a diverse kitchen team in a fast-paced environment, staying calm and efficient through high-pressure peak periods.
- Prioritised tasks and communicated clearly between kitchen and front-of-house teams to meet tight deadlines.

CERTIFICATES

White Card – General Construction Induction

KNOWLEDGE & PROJECTS

Smart Mine AI – Conceptual Design Project **[2026 – Present]**
Collaborative Project with Professor Erkan Topal, *Curtin University, 2026*

- **Project Goal:** Designing an AI-powered ore sorting and grinding optimisation system for the run-of-mine value chain, to reduce the energy, water and money spent on grinding low-value rock.
- **AI Design:** Four-stage pipeline: YOLO11 ore/waste detection, segmentation (YOLO, SAM, OpenCV) for particle size and shape, machine-learning grade prediction (Random Forest, gradient-boosted trees, neural networks) fusing camera, hyperspectral and XRF data, and model-predictive mill control.
- **Prototyping & Automation:** Built Python simulations and a proof-of-concept operator dashboard and defined a digital twin and MLOps platform for continuous model improvement.
- **Key Results:** In simulation, sensor fusion cut metal lost to rejects from 0.41% to 0.19%, and feedforward control reduced product-size variability by about 78% (idealised case). Results use synthetic data and are targets for a pilot.

Skin Check Champions – Capstone Client Project **[2026]**
Backend Developer & AI Quiz Assessment, *University of Wollongong, 2026*

- **Backend Development:** Built core backend services, APIs and data flow for user interactions and quiz features.
- **AI Quiz Module:** Developed an AI-powered quiz system that gives personalised learning feedback, turning AI into a practical product feature.
- **Client Collaboration:** Worked directly with an external client, presenting updates and acting on their feedback.
- **Team Contribution:** Collaborated in a six-member agile team on requirements, planning and delivery against deadlines.

References will be provided upon request.